

Sequence labelling and error analysis for code-mixed text by an averaged perceptron approach

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ABSTRACT

In mixed-language settings, NLP models face challenges because of intertwined grammatical structures and language discrepancies. This paper analyses a Part-of-Speech tagger used on Hindi-English code-mixed (Hinglish) data and compares it with monolingual English and Hindi data. The model achieved an accuracy of 93.60% on the English dataset, accuracy of 95.95% on the Hindi dataset, and 94.89% on the Hinglish dataset, respectively. This showed strong model performance in both monolingual and code-mixed settings. The study also tried to understand the patterns in sentence structure that led to tagging errors. It was found that the most misclassifications occurred between nouns, verbs, proper nouns, and adjectives because of their overlapping lexical forms. The analysis of sentence length indicated that medium-length sentences yielded marginally higher accuracy than long sentences. In contrast, the error distribution per sentence showed that most sentences contain zero or one tagging error, with a cascading effect observed in a minority of cases where a single error in a sentence led to more errors and isolated errors. In the future, researchers can focus on understanding the reason for dual language error patterns, such as common misclassifications in a complex sentence.

Keywords: *NLP, Parts of Speech, POS tagger, mixed languages*

INTRODUCTION

The most fundamental task in Natural Language Processing (NLP) is Part-of-Speech (POS) tagging. Accurate POS tagging is important because many NLP tasks, includes machine translation, information extraction, and entity recognition, depend on reliable grammatical analysis. In POS tagging, a grammatical label, such as noun, verb, adjective, or adverb, is tagged to each word in a sentence. These labels help machines understand the basic structure of a sentence and how words relate to one another.

POS tagging systems are generally designed for single-language text. However, in real-world settings where multilingual communities exist. For example, in India, native speakers often combine words from multiple languages like Kannada-English, Hindi-English, and Malayalam-English within the same sentence. It is found that POS taggers struggle with code-mixed language sentences. In India, English is frequently used in everyday communication, especially in informal settings such as conversations on

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social media platforms, e.g., Twitter and Facebook. It is found that speakers tweet by combining words from both languages while maintaining a single conversational flow. The users write Hindi words using the Roman English alphabet for communication.

The research paper primarily focuses on the challenges faced in POS tagging in code-mixed data from Hindi and English. The English language generally uses a Subject–Verb–Object (SVO) word order, while Hindi typically follows a Subject–Object–Verb (SOV) structure. Therefore, POS tagging becomes complicated when these two patterns appear together in code-mixed sentences. In social media conversations, we can find not only mixed language usage for communication but also spelling variations, abbreviations, and informal expressions while posting messages on different social media platforms. It is imperative to investigate the performance of a POS tagging system applied to Hindi–English code-mixed data collected from social media. In this paper, instead of focusing only on overall accuracy, the paper shows analyses of the types of tagging errors produced by the model and the conditions associated with these errors.

This study aims to better understand the linguistic factors, such as common misclassifications, sentence structure, and contextual influences. These factors were included because they affect tagging performance in mixed-language text, and the paper seeks to understand the nature of these errors, which would be important for improving NLP systems operating in multilingual environments.

This research is important because code-mixing is prevalent in social media, informal communication, and everyday speech in multilingual communities. Standard NLP tools trained on monolingual data perform poorly on code-mixed text. Understanding error patterns helps improve POS taggers for real-world multilingual applications.

LITERATURE REVIEW

Several studies have explored POS tagging for different languages.

Prevalence of code-mixing trend on social platforms

Danet and Herring's 2007 research examines how multilingual individuals navigate linguistic choices across different online settings, including studies of Egyptian online language preferences, Swiss citizens' multilingual practices, code-switching in German forums, and Swedish speakers' digital communication, with one theory suggesting English serves as a neutral lingua franca that transcends political and cultural divisions in multilingual societies (3).

Extent of Hindi-English mixing code

Vyas et al.'s 2014 study describes initial efforts to create a multi-level annotated dataset of Hindi-English code-mixed text, collated from social media forums. It explores language identification, back-translation, and normalisation. They demonstrated that language identification and transliteration for Hindi are two

major challenges that significantly impact POS tagging accuracy. The study revealed that while there are easily identifiable cases of borrowing and mixing at the two ends, a large majority of words form a continuum in the middle, emphasising the need for automatic processing of the data. (10)

Si's 2010 study examines code-switching patterns in Bollywood movies from the 1980s, 1990s, and 2000s through lexical transcripts of dialogues between characters of equivalent age and socio-economic status in India over the three decades. The seven movie dialogues showed a massive increase in overall English usage over this period, particularly among young speakers, while the complexity of code-switching also increased, with clause-boundary alternations becoming more frequent at the expense of single-word insertions. (9)

Natural Language Processing of Hindi–English Code-Mixed Text

Singh, Sen, and Kumaraguru's 2018 study addresses the growing prevalence of code-mixing on social media platforms by presenting a unique language-tagged and POS-tagged dataset of English-Hindi tweets related to five significant incidents in India that generated substantial Twitter activity. The dataset is distinctive in that it resembles typical real-world tweets and showcases the critical gap in annotated training data necessary for automatic Parts-of-Speech tagging in multilingual societies. The tweets were manually annotated by bilingual experts, adding specialised categories for proper nouns, negative particles, and interrogative pronouns. They have analysed the errors and highlighted difficulties at code-switch boundaries and in the adjective–adverb category due to grammatical differences between Hindi and English. (6)

Petrov, Das, and McDonald (8) proposed a universal part-of-speech (POS) tag set consisting of twelve coarse grammatical categories to standardise linguistic annotation across languages. Their experiments demonstrated its usefulness in multilingual tagging, unsupervised grammar induction, and cross-language dependency parsing. This universal tag set enables more meaningful cross-linguistic comparison and improves the transferability of NLP models between languages. (8)

Xu, Ji, Li, and Lu's 2023 review addresses the critical challenge of small data by analysing the limitations imposed by data scarcity and presenting comprehensive strategies across multiple levels to overcome these constraints. The authors systematically examine methods at the data source level, including data extraction from publications, materials database construction, and high-throughput computations and experiments; at the algorithm level through modelling algorithms optimised for small data and imbalanced learning techniques; and at the machine learning strategy level via active learning and transfer learning approaches. The review highlights that support vector machines and Gaussian process regression are particularly well-suited to low-dimensional, small-data regression tasks. At the same time, ensemble methods like random forests can improve prediction accuracy by integrating multiple algorithms. Feature engineering remains integral to successful small-data modelling through careful feature selection and dimensional reduction. (11)

METHODOLOGY

Dataset

The dataset used was entitled “A Twitter Hindi-English Code-Mixed Dataset for POS Tagging” and contains 1,489 individual tweets scraped from X (Twitter) with a total of 33,010 tokens. [7] As the dataset used was created by Singh et al. described that, these tweets were filtered to ensure that they all adhere to the following criteria.[6]

- these sentences have colloquial usage of the languages,
- they contain a mix of formal and informal grammatical structure, and

Each tweet contains

at least three Hindi tokens

– at least three English tokens

– at least two contiguous Hindi tokens, and– at least two contiguous Hindi tokens, and– at least two contiguous English tokens. As described by Singh et al., this can be explained by the misidentification of languages of named entities, which are left in the dataset. [6] These sentences are related to significant news events that resulted in active discourse on social media.

Despite this filtering, tweets that are wholly English or wholly Hindi are present in the dataset, as shown in Table I.

Attribute	Number
Code-Mixed Tweets	1077
English Tweets	343
Hindi Tweets	69
Total Tweets	1489
Training Tweets	1189
Testing Tweets	300
Tokens	33010

Table 1: Distribution of tweets according to languages, training and testing data, and the total number of tweets and tokens present in the data.

The dataset presents the model with two distinct syntactic systems: English typically follows a Subject–Verb–Object (SVO) word order, while Hindi commonly follows a Subject–Object–Verb (SOV) structure. The presence of these differing grammatical patterns allows the model to encounter structural variability similar to that found in real-world multilingual communication, whereas in code each word has been manually tagged according to language ('hi,' for hindi 'en, for english' 'CM') and part of speech (ADJ for adjective, ADP, ADV for adverb, CONJ for conjunction, DET, NOUN, PART, PRON for pronoun, PROPEN for proper noun, VERB). Hashtags and user mentions were left in. These, as well as named identities (those not mistagged), sub-lexically code mixed tokens, acronyms, and symbols (including punctuation) were tagged as rest in the language column. These POS tags follow the conventions of the universal tagset outlined by Petrov, Das, and McDonald (2011). [8] Additionally, the tags described subsequently have been used. PART_NEG represents negative particles, PRON_WH represents interrogative pronouns, and X denotes foreign words, typos, abbreviations, and punctuation.

Both source corpora are distributed in CoNLL-U format, processed using a Python-based parsing pipeline that extracts token and tag information while discarding metadata lines. Following preprocessing, the data was converted into a list of sentences, each represented as a list of (word, POS) tuples. From the Python library "sklearn.model_selection", the "train_test_split" function was used to randomly sample tweets from throughout the dataset in an approximate 80:20 ratio. This resulted in 1,189 tweets is being present in the training list and 300 tweets in the testing set, as shown in Table I. These two lists were used for subsequent tests. A randomly chosen sentence and its corresponding tags are shown in Table II.

Token	Lang	Actual Pos	Predicted Pos
Waiting	en	VERB	VERB
.	rest	X	X
Just	en	ADV	ADV
asking	en	VERB	VERB
to	en	PART	PART
eat	en	VERB	VERB
one	en	NUM	NUM
of	en	ADP	ADP
us	en	DET	DET

Mere desh	rest	PROPN	PROPN
ki	rest	ADP	ADP
awaz	rest	PROPN	PROPN
voices	en	NOUN	NOUN
.	rest	X	X

Table 2: A sample tweet from the dataset.

This representation enabled efficient processing, and the dataset was then partitioned into training and testing subsets using a randomised 80:20 split. The training subset was used to learn model parameters, while the testing subset was reserved for evaluating the model’s performance on previously unseen sentences.

Model

The model implemented in this experiment is the averaged perceptron Part-Of-Speech tagger. Matthew Honnibal (4) mentions that the perceptron is a linear classification algorithm that predicts the class of an input token based on a weighted sum of its feature values. The perceptron generates a feature vector for each token in a sentence to represent the word’s lexical, structural, and contextual properties. The perceptron also assigns scores for all possible part-of-speech tags by summing the weights of all active features. During training, weight updates change the model’s parameters to optimise its performance at making these predictions. The model suggests a tag for every token. The suggestion was compared to the ground-truth annotation. If the predicted tag differed from the actual tag, then the weights for the correct tag were increased while the weights for the incorrectly predicted tag were decreased. Repeatedly, the model incrementally learnt from the training examples about the relationship between features and part-of-speech categories.

The model preserved cumulative weight totals for stability and did not rely solely on the most recently updated weights. During training, the final weights were calculated as the average of these accumulated values.

This averaging process helps reduce inconsistencies during training and, in turn, improves the model’s ability to generalise to unseen data. We defined four categories: true positive (TP), false positive (FP), true negative (TN), and false negative (FN), using a confusion matrix, where the columns are the true values and the rows are the predicted values. In these categories, “positive” and negative” indicated whether the prediction was positive or negative, while true” and false” indicated whether the prediction matches the true value. The precision measures the proportion of correct predictions the model makes. The model also used a feature-extraction process that incorporates several types of linguistic information. These include lexical features such as the current token, contextual features such as the surrounding

words in the sentence, and morphological features such as prefixes and suffixes derived from the token. Additionally, the model considered the predicted tags of the two preceding tokens in the sequence. By incorporating these sequential dependencies, the model can capture local syntactic patterns within sentences.

The model was trained on approximately 80% of the available dataset, with the remaining 20% used as a test set. The model was trained for five full passes (epochs) over the training data, with the training sentences shuffled before each pass. A fixed random seed (42) was used for corpus shuffling. At inference time, the model performs greedy left-to-right sequential decoding, in which the two most recently predicted tags are retained as decoding history and used as input to the feature extractor for subsequent tokens. Training was conducted over multiple iterations, during which the training sentences were randomly shuffled. Following training, the model's performance was evaluated on the test dataset. Evaluation involved comparing the predicted tags with the true annotations for each token. From these predictions, several performance metrics were calculated as summarised in Table I. The training process utilised the `train()` function of the Perceptron, which takes lists of word-tag pairs as input. When used to tag a set of tokens, this tagger leverages its predictions for surrounding words to influence the prediction for the current word. As shown in Table II, Following training, the model's accuracy was evaluated on the test data. This evaluation also involved analysing the different types of errors encountered in the model's predictions.

Accuracy was measured as the proportion of tokens for which the predicted tag matched the ground-truth annotation. In addition to overall accuracy, precision and recall were calculated for each tag category. Precision represents the proportion of predicted instances of a tag that are correct, defined as:

$$\text{Precision} = \text{TP} / (\text{TP} + \text{FP})$$

Recall represents the proportion of true instances of a tag that are correctly identified by the model, defined as:

$$\text{Recall} = \text{TP} / (\text{TP} + \text{FN})$$

These metrics allowed a detailed evaluation of model performance across different grammatical categories. In addition to these metrics, further analysis was conducted using a confusion matrix, which records the frequency with which each true tag is predicted as another tag. This analysis provides insight into systematic misclassification patterns within the model's predictions. Sentence-level statistics were also examined to determine whether tagging accuracy varied with sentence length or syntactic complexity.

RESULTS AND DISCUSSION

Accuracy, Precision and Recall

The performance of the POS tagging model was evaluated using accuracy, precision, and recall. Accuracy measures the percentage of words the model tagged correctly.

The model achieved 93.60% accuracy on the English dataset, 95.95% on the Hindi dataset, and 94.89% on the Hinglish (Hindi-English code-mixed) dataset. This means that only a small percentage of predictions were incorrect. Compared to the models on average, it is just 1.06% less accurate than the Hindi model and 1.3% more accurate than the English model. Overall, the results show that the model performs well across both monolingual and code-mixed language data. In addition to the differing grammatical structures of the two languages, the fact that this data was sourced from social media means that the grammar, spelling, and sentence structure pose a challenge for the tagger. Additionally, the inconsistent spellings arising from the informality and Romanisation of the data (where Hindi words are written in the Roman script rather than Devanagari) make this task challenging for monolingual models. These complications and inconsistencies across the data likely affect the Hindi-English Tagger as well.

Among the three datasets, the Hindi dataset produced the highest accuracy. One possible reason is that the grammatical structure in Hindi sentences within the dataset is relatively consistent, which helps the model identify patterns more easily. The Hinglish dataset, which contains mixed Hindi and English within the same sentence, is usually more difficult to process. Despite this, the model still achieved high accuracy, suggesting it can learn patterns across multiple languages.

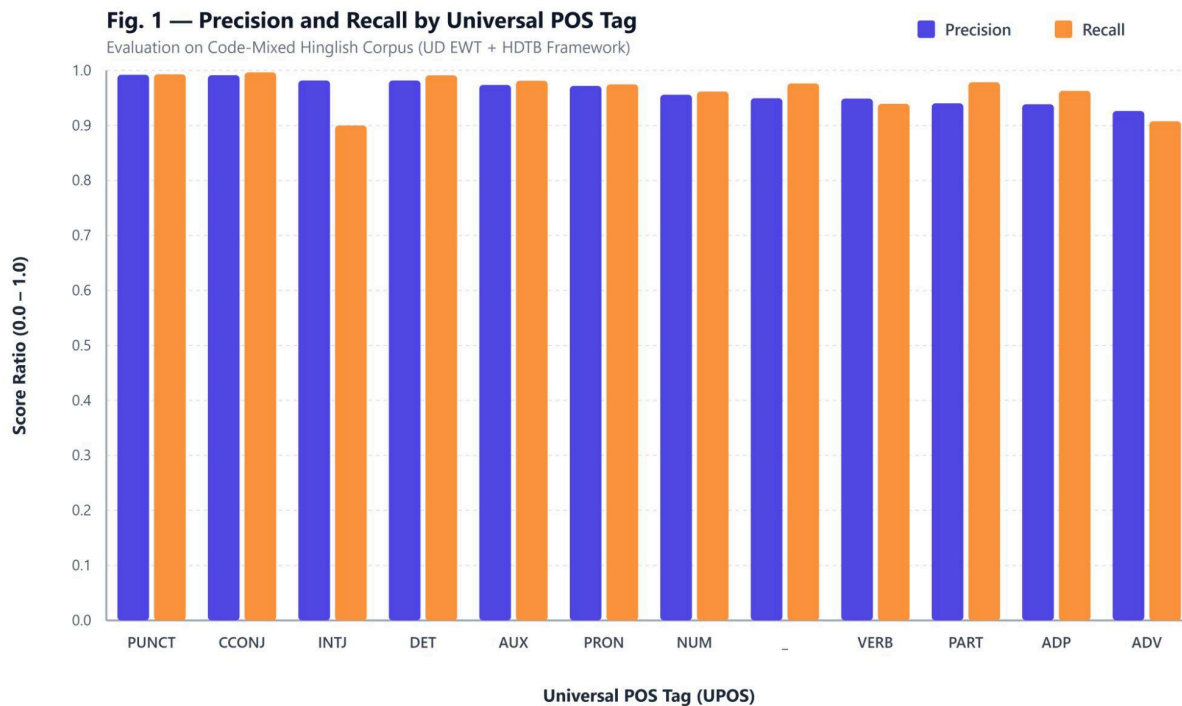


Fig 1: Precision and Recall for different POS tags

Figure 1 shows the precision and recall of the model, analysed for individual POS tags. Most POS categories, such as CCONJ, PRON, ADP, AUX, and DET, exhibited precision and recall values close to 1.0, indicating highly reliable classification. However, certain tags, such as INTJ and SCONJ, showed slightly lower recall, suggesting these categories were more difficult for the model to identify consistently. This indicates that the model is very reliable when identifying these parts of speech. A possible reason for this could be that certain words can function as more than one part of speech depending on context. For example, a word may appear as a noun in one sentence but as a verb in another. Because of this ambiguity, the model may occasionally misclassify the tag.

Another challenge arises from the informal nature of social media text, in which many datasets are collected. Code-mixed text often contains inconsistent spelling, abbreviations, and informal grammar. Additionally, Hindi words are frequently written using Roman script rather than Devanagari, which adds further complexity to the tagging task.

Understanding Prediction Errors

Certain POS tags are frequently confused with other textual contexts. Two perspectives can be used to interpret the confusion matrix:

1. For each incorrectly predicted POS, what was the actual POS?

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- For each POS, what categories does the model most often predict instead?
- Analysis of the confusion matrix in Fig. 2 reveals several systematic misclassification patterns. The heatmap shows that certain POS categories are more frequently confused with others. Most notably:
- Nouns (NOUN) and Verbs (VERB) are occasionally confused with each other, likely due to the differences in grammatical structure between Hindi (Subject–Object–Verb) and English (Subject–Verb–Object).
- Adjectives (ADJ) and Adverbs (ADV) show some confusion, which may stem from their similar morphological features in code-mixed text.
- Pronouns (PRON) are generally well-classified with minimal confusion with other categories.
- Proper nouns (PROPN) and regular nouns (NOUN) show some overlap, which is expected as their distinction can be subtle in informal social media text.
- Most errors are isolated, with a low frequency of cascading errors, suggesting that misclassification of one token does not typically lead to widespread errors in subsequent tokens.
- The modal number of errors per sentence is 2, indicating that while errors do occur, they are not pervasive throughout the sentences.

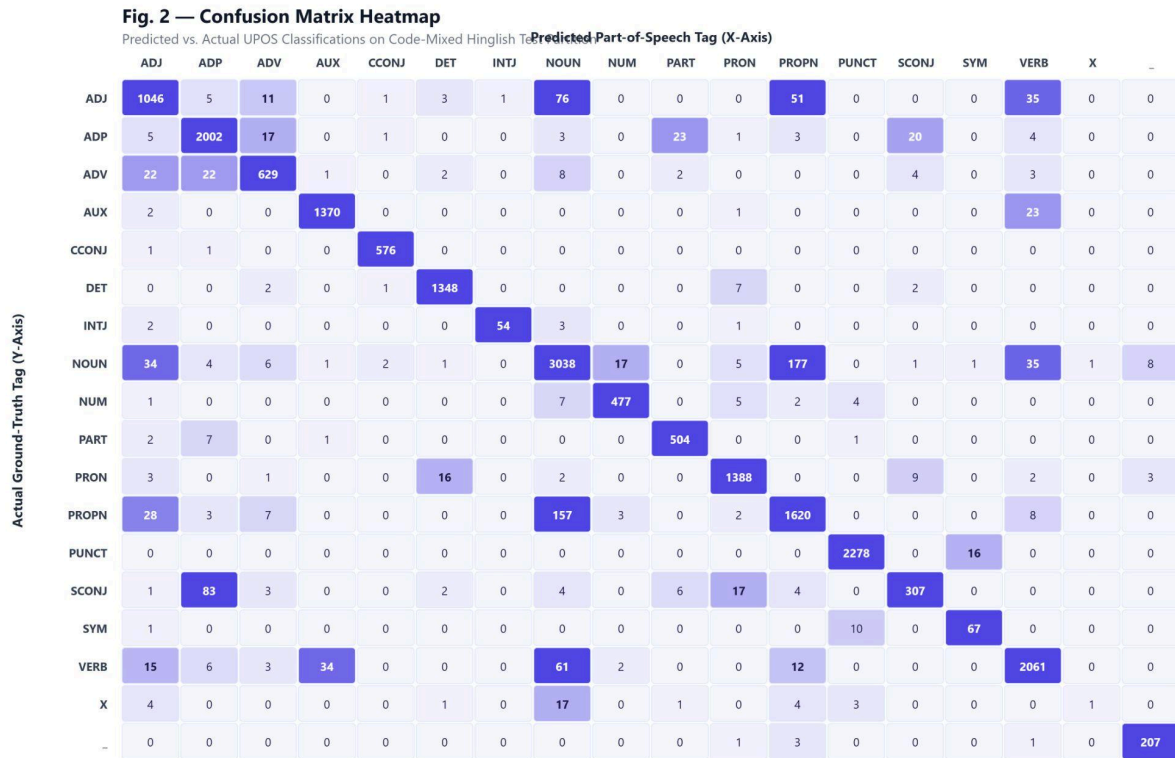


Figure 2: Heatmap of prediction errors organised according to actual and predicted POS

Sentence Length and Model Accuracy

An important factor in assessing model performance is understanding how accuracy varies with sentence characteristics. Longer sentences or more complex syntactic structures might present additional challenges for the POS tagger. An analysis was conducted to examine the relationship between sentence length and tagging accuracy.

The results of this analysis show that tagging accuracy remains relatively stable across different sentence lengths, though with some minor variations. Short sentences (typically 5–10 words) show very high accuracy, as they usually contain simpler grammatical structures. Medium-length sentences (11–25 words) maintain good accuracy, though with slightly more variation due to increased complexity. Longer sentences (26+ words) present more challenges, as they may contain more complex syntactic dependencies and a greater variety of POS categories. However, the model still performs reasonably well on these sentences.

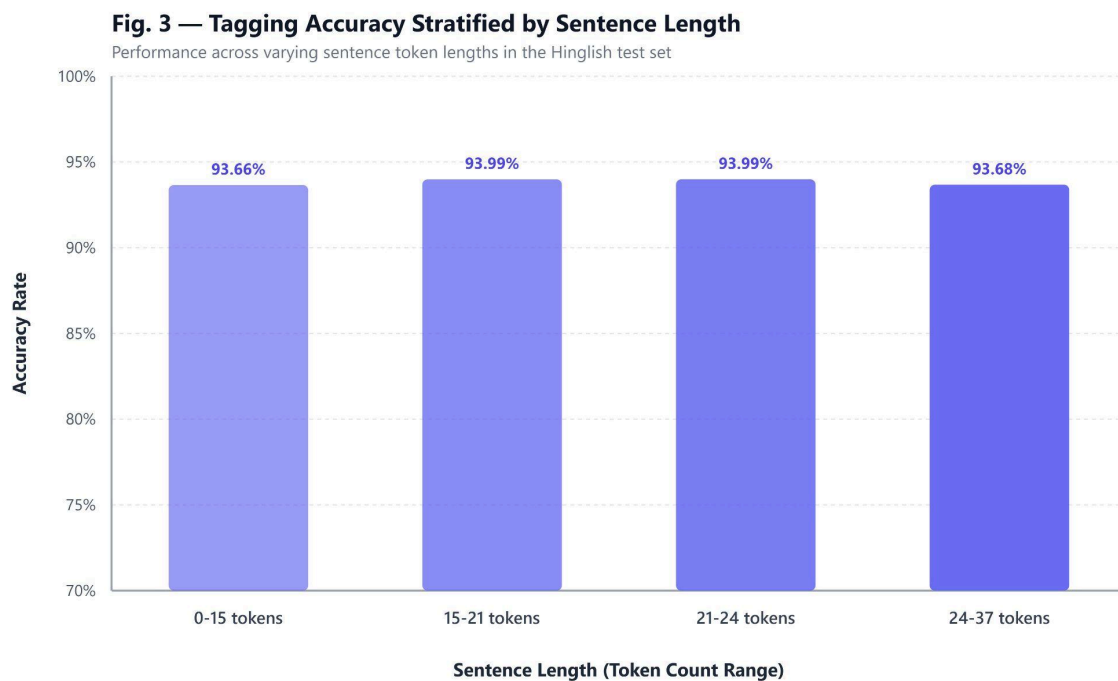


Figure 3: Accuracy by Sentence Length

This analysis demonstrated that while sentence length does impact accuracy to some degree, the effect is not dramatic. The model's ability to maintain reasonable performance even on longer sentences suggests that the Averaged Perceptron with feature engineering is reasonably robust to syntactic complexity. However, there is room for improvement through more advanced architectures such as neural networks or transformer-based models that might better capture long-range dependencies.

Error Distribution per Sentence

Most sentences had zero or one tagging error, which indicated that the model typically predicts most tokens correctly. However, some sentences contained multiple tagging errors. This may occur because the tagging algorithm considers the tags of nearby words when making a prediction. If one token is tagged incorrectly, it may affect the predicted tags of surrounding tokens, potentially causing mistakes within the same sentence. Figure 4 shows that the most common error count in a sentence is 0, with a sharp decrease in the number of sentences as the error count increases beyond that.

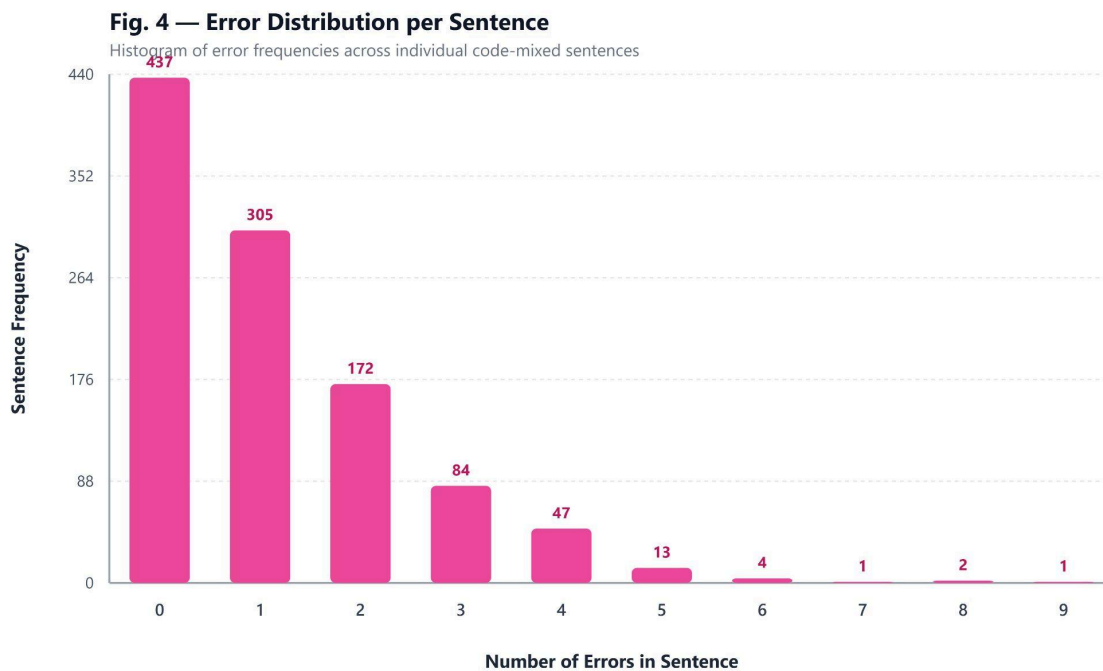


Figure 4: Error distribution per sentence

LIMITATIONS

This study aims to understand the different types of errors affecting POS tagging in Hindi-English code-mixed text. The tagger has achieved accuracies of 93.60%, 95.95%, and 94.89% on the English, Hindi, and Hinglish datasets, respectively, which suggests a strong ability to tag parts of speech correctly. There are limitations in the work that have affected model performance in capturing the prior error analysis.

Data Size

Although the model demonstrates high accuracy across all three datasets, the size of the dataset used for training and testing still limits the model's performance. Studies suggest that larger datasets can

significantly improve POS tagging performance, particularly for code-mixed language data. Smaller data samples increase the risk of overfitting, underfitting, and class imbalance (11). The lower volume of Hinglish training data, in particular, may explain the difficulty the model encounters in consistently distinguishing between closely related categories such as NOUN, VERB, PROP, and ADJ, which together account for the largest share of misclassifications observed in the confusion matrix

Data Homogeneity

Another limitation comes from the source and nature of the data used to build the model. As the dataset was taken from social media, the sentences reflect informal conversational language that may not be representative of the full spectrum of code-mixed Hindi-English usage. Consequently, the model struggles to generalise to code-mixed text from other domains, such as news articles, academic writing, or spoken transcripts.

Sentence Length Sensitivity

The analysis revealed that very short sentences, those with fewer than 15 words, yielded the lowest accuracy among all the sentence length groups. This suggests that the model relies significantly on surrounding word context to make accurate predictions, and that the reduced contextual information available in shorter sentences introduces greater ambiguity. Similarly, sentences exceeding 24 words showed a decline in accuracy, indicating that longer and more structurally complex sentences pose an additional challenge.

Cascading Errors

The error distribution analysis demonstrated that POS tagging errors tend to have a cascading effect, whereby a single misclassification influences the predicted tags of neighbouring words, resulting in clusters of errors within the same sentence. Although the modal error count per sentence was low, this cascading behaviour indicates a structural limitation in the model's sequential decision-making.

CONCLUSION

This study evaluated the performance of a POS tagging model on Hindi-English code-mixed datasets, with a particular focus on understanding the types and distributions of prediction errors. The model demonstrated consistently high accuracy across all three datasets, achieving 93.60% on English, 95.95% on Hindi, and 94.89% on Hindi -English models, indicating that it is capable of understanding cross-lingual patterns even in informal, code-mixed scenarios. The error analysis identified several common misclassifications that lie between the NOUN, VERB, PROP, and ADJ categories. These tags are particularly susceptible to confusion due to their overlapping lexical forms and context sensitivity. The structural contrast between English (SVO) and Hindi (SOV) grammatical orders further exacerbates this challenge, as verbs occupy different positions in both languages. This makes it harder for the model to learn stable positional indications. The informal nature of social media text adds another layer of complexity due to spellings and Roman-script representations of Hindi. It was found

that medium-length sentences, in the 21–24-word range, yielded the highest accuracy. The greater structural complexity was found in very short sentences because they lack sufficient grammatical context. Notably, the differences across sentence length groups were small, suggesting that the model generalises reasonably well across varying sentence structures. The error distribution analysis showed that the majority of sentences contained zero or one tagging error, reflecting the model’s overall reliability. However, a cascading effect was observed in some sentences, where initial misclassifications influenced the tagging of adjacent words, resulting in clusters of errors. Future work can focus on expanding the training corpus to include a wider range of code-mixed usage patterns and dialects. Incorporating pre-training on large monolingual corpora, alongside fine-tuning on code-mixed data, could strengthen the model’s language-specific representations.

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